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Design and analysis of double girder overhead crane system

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ABSTRACT

Cranes, which are very useful in function, are widely used by manufacturers producing finished or semi-finished products that need to be transported in shipyards, in the aviation industry, due to the advantages of their high carrying capacity and different movement variations. Within the scope of this study, it covers the works carried out for the crane system designed for the transport of galvanized coils with a weight of at least 2 to a maximum of 6 tons, 0.20–5 mm thick, with a maximum outer diameter of coils up to 1500 mm. First of all, the elements that make up the crane system were determined and theoretical calculations were made. Then solid models were created. The design and assembly cases were created according to the situations where the designed car group and the load are located at the bridgehead and in the middle. In this way; The effects of load, carrier frame, bridge and car were investigated. The conditions of these two conditions, which were determined while carrying out these examinations, with and without the carrier frame were also examined.The results obtained as a result of 10tons applied in beams and rope were below the safety values. As a result, a work that can be manufactured has emerged as a result of the analysis applied to the crane designed for the needs.

1. Introduction

Since the earliest known humans, history has always written: From the very beginning, people have tried to make inventions that could make their work easier. The development of technology and the fact that people are now conscious of some things have forced them to make magnificent inventions at the points where their own power is not enough. One of these inventions is cranes.

In the production industry, it is often not possible to transport semi-finished or finished products in production areas with manpower due to the weight of the product or due to occupational safety principles. In order to lift these products and move them to another place, special vehicles called cranes are used to move the related products by turning, ascending and descending or horizontal axes, depending on the situation. Cranes, which are very useful in function, are also widely used in the aviation industry due to their high carrying capacity and the advantages of different movement variations. In general, it is possible to divide cranes into two basic classes according to their mobility and lifting capacities. Cranes can be divided into six main sections according to their mobility.

- Fixed cranes that are stationary in one place,
- Crawler cranes moving on a pallet,
- Rubber-tyred cranes used as mounted on rubber-tyred vehicles,
- Gantry cranes, which are generally used in open areas (port, mine, etc.),
- Gantry cranes moving on rails and
- Tower cranes that are used vertically and can be quite high

Additionally; Cranes can also be classified as single girder cranes and double girder cranes according to their beam structure. The image of the single girder cranes are shared in Fig. 1 below.

In the shipbuilding industry and other factories, overhead cranes are mostly used. This type of overhead cranes can be of single-girder or double-girder structure, depending on the required carrying capacity and the characteristics of the building in which they will be installed. Although there are some structural differences, the working principles of both types of cranes are basically the same. The structural components of a standard overhead crane are shown in Fig. 2.

The basic components of overhead traveling cranes consist of the walkway and the bridge moving on it. On the bridge, it consists of the moving trolley and the winding pulley system connected to the trolley. The hook system is hand control controlled (Öğün, 2014). Cranes are purely dynamic systems; oscillation, accelerated movements, etc. that occur during carrying the load. Dynamic loads caused by these reasons create a force directly on the load to be carried. For this reason, it can cause problems such as disconnection from connection points, and as a result, it causes work safety problems or material damages. Recently,

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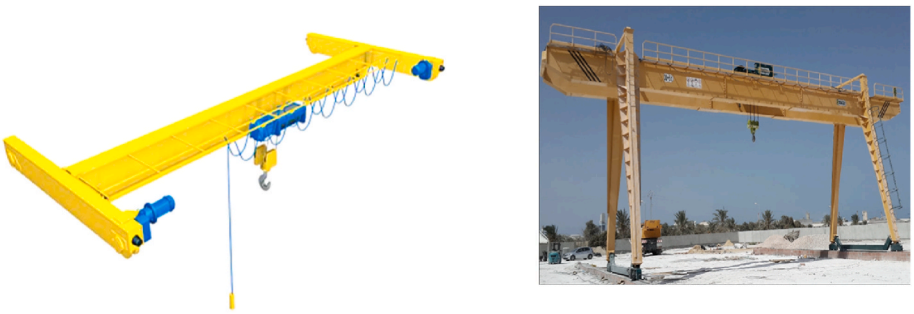


Fig. 1. Single girder overhead Crane.

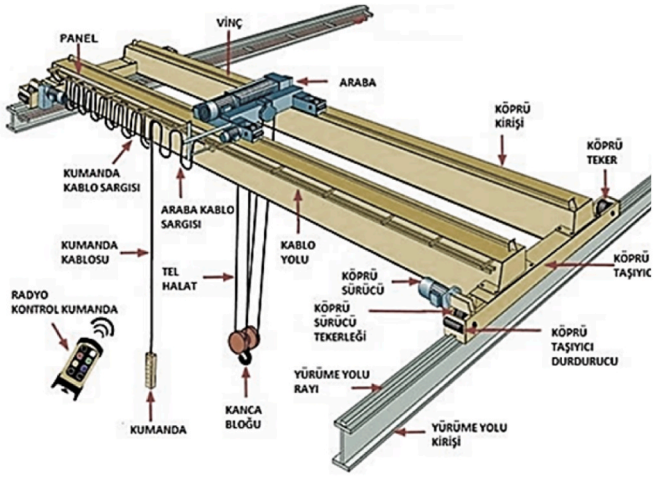


Fig. 2. Structural components of a standard overhead crane.

crane manufacturers have added functions such as soft start and soft stop to their products with new generation controllers that will not create additional force on the load caused by these operations during lifting, carrying and lowering, and have offered them to the use of the industry. In addition to the technological features of the cranes, the possible risks of crane operators using cranes in accordance with lifting standards and guidelines are reduced. However, despite all these precautions and high-tech applications, possible risks could not be completely eliminated. In these cases, it is important that the apparatus used by hanging on the hook of the crane are strong enough to meet the dynamic forces.

1.1. Working principle of overhead crane systems

1.1.1. Drive unit of overhead crane

The drive unit is the power device that drives the operating mechanism. Electric drive, internal combustion engine drive, human drive, etc. in the general drive gear. electricity is clean and economical energy, electric drive is the main propulsion method of modern cranes.

Table 1
Main lift group account.

Lifting Capacity Qy	10.000 kg (Rope load)
Lifting Speed Vk	0–15 m/min (stepless)
Equipment t	8/4
Lift height	5 m
Lifting speed	4 m/min
Bridge Travel Length	6 m
Bridge Motion Angular Speed	147,7 rpm
Bridge motion motor power	0,37 kW
Car Motion Angular Speed	16 rpm
Car motion motor power	0,25 kW
Steel rope	8 mm
Hook	DIN15401
Material	St 37

1.1.2. Working mechanism of overhead crane

The working mechanism of overhead crane includes: lifting mechanism and operating mechanism.

- The lifting mechanism is to achieve the vertical lifting mechanism of the object, so it is the most important, the most basic mechanism of the crane.
- The working mechanism is the mechanism for transporting objects horizontally by means of a crane or a lifting trolley, which can be divided into rail operation and trackless operation.

Pick-up device for overhead cranes The pick-up device is a device that attaches the object to the crane by means of a hook. Depending on the type, shape and size of the suspended object, different types of collection devices are used.

The aim of this study is to design a crane for the transportation of

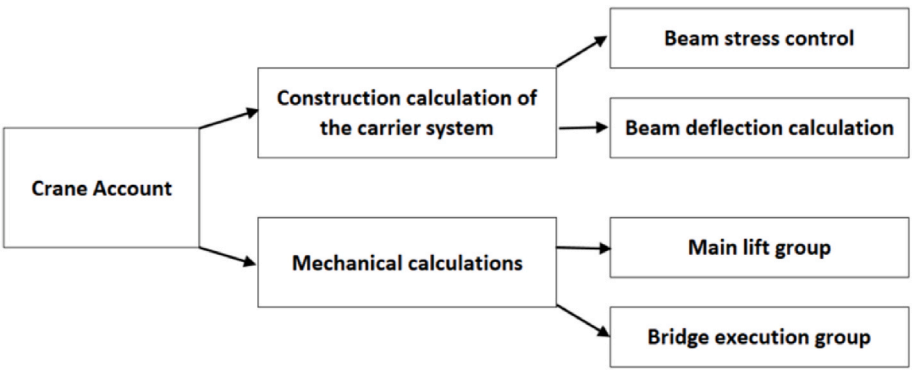


Fig. 3. Crane design stages.

Table 2
Materials determined as a result of calculations for lifting appliances.

Selected rope	dh = Ø 12 mm (6x36WS TS 1918/14)
Selected drum diameter	Ø 400 mm
Selected reel diameter	Ø 500 mm
Selected engine for car drive	4 kW 1000 rpm
For carriage Selected wheel diameter	Ø 15 cm
Selected motor for Bridge Drive	10 kW 1000 rpm
Selected gearbox	SEW FA87 AD5
Selected coupling	1.640 Nm (Number of gearboxes: 4, Sf = 2.6)
Selected wheel diameter	Ø 40 cm
Wheel diameter selected for the swivel group	Ø 25 cm
Selected engine in Lift Engine Calculation	65 kW 1000 rpm

Table 3
Strength values of the designed crane and rope.

For the crane;		For the rope	
Yield Strength	Min. 235–340 MPa	Tensile Strength	Max. 333.92 kg/cm^2
Tensile Strength	360–510 MPa		

steel sheet plates with a width of 2.7m and a weight of 3500 kg in the iron and steel industry.

1.1.3. Overhead crane control system

It mainly acts through the electrical system control to manipulate the entire movement of the crane mechanism to perform various operations.

Most overhead cranes begin a vertical or vertical horizontal working stroke after receiving the lifting device, unloading it when it reaches its destination, and then unloading the stroke to the receiving ground to complete one working cycle, and then performing a second lift. Generally, lifting machines operate by sequential material removal, displacement and discharge, while the corresponding mechanism works intermittently. Lifting machines are mainly used to transport solid goods, equipped with ladles to move loose materials such as coal, ores and grain, and buckets to lift liquid materials such as steel. Some lifting machines, such as the elevator, can also be used to transport people. In some cases, for example in ports and stations where the loading and unloading of materials is the main operating machine, the lifting equipment is also the main operating machine.

2. Theoretical calculation, design and analysis

2.1. Theoretical calculation stage

While calculating the crane, attention was paid to the working stages shared in Fig. 3.

Calculations were made by loading S_L, S_H and S_W as distributed loads on the beam.

SG = 186(ton) x 9.81(m/s2) = 1824.660 x 1.14 = 2080112 N

S_L = 10(ton) x 9.81(m/s^2) = 98.1 x 1.15 x 1.14 = 128609 N

S_H = 10x(SL)+10xArabanın ağırlığı x 9.81(m/s^2) = 49.05 x 1.14 = 55917 N

S_W = 147233 N

While designing the crane system examined in this study, it is necessary to know the design parameters such as the lifting capacity of the crane system, the lifting height, the bridge span and the bridge travel length as a reference. The design parameters and operating speeds of the Crane system determined (and calculated) according to the physical conditions are shared in Tables 1 and 2.

2.2. Design stage

The designs are designed in accordance with the specification within the scope of crane standards (TS EN 13001-3-2, 2014). Crane was designed using the Solidworks program according to the calculated theoretical parameters.

A carrier frame is designed for the Crane bridge assembly. In the crane system, which is designed in such a way that the lifting group is combined with the load, the effect of structurally on the load is examined. To this end; The crane system was requested to be evaluated in two different ways, both with the carrier frame and without the carrier frame.

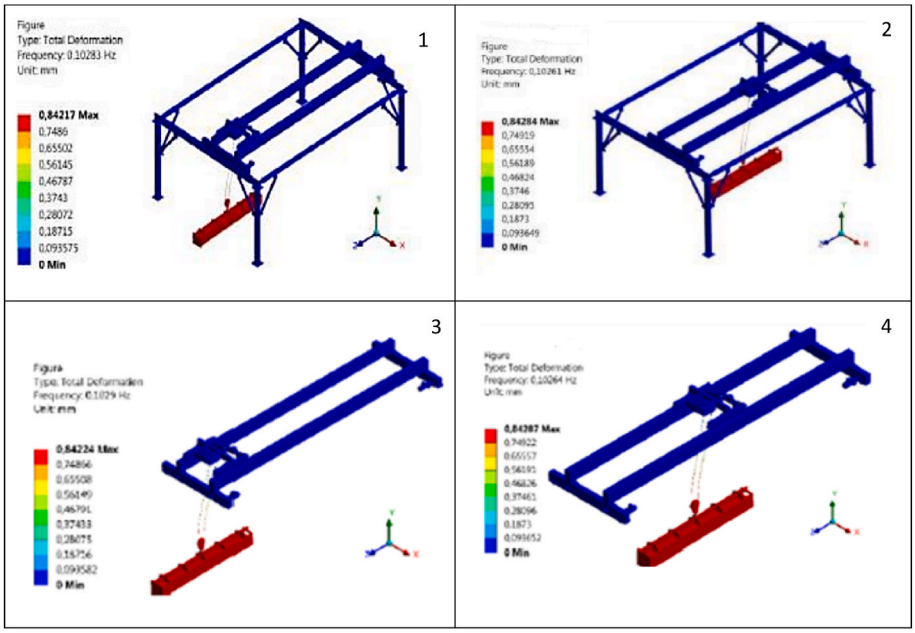


Fig. 4. Bridgehead and mid-bridge points applied and analyzed.

2.3. Analysis stage

The parts drawn using the Solidworks program were created in the same program as the assembly of the product. Crane assembly models were transferred to the ANSYS Workbench program. There may be loads, moment loads or different forces that may come to the part in the design (Akkurt, 1990; Erdil, 2007). Analysis can be made for different points where the load can come, but mostly the points where the maximum load will be determined with the basic principles and the analyzes are made on these points. The designed crane will be operated in the factory indoor environment and therefore the wind load has been neglected (Mete, 2010). In the designed Crane system, two situation conditions (pertaining to the regions where the maximum load is important) were determined and the effect of the structural load on the places was examined.

The first condition is the presence of the car group and the load at the bridgehead and includes its inspection accordingly. The second condition is that the car group and the load are in the middle of the bridge, and it includes its examination accordingly.

The strength values of the designed crane material and the selected rope are shared in Table 3 below.

After ensuring that the stress distribution on the model under various loads remains below the desired limit values, it is very important for engineering problems to determine the minimum weight and the most suitable design to carry the existing loads (Güder, 2014; TS EN 12999, Aralık 2005, & Vinç ve Yükleyici Vinçler Yıldız, 2006, pp. 10–33; Okuda, 2023). In the study; After the product design, material selection, boundary conditions and loading conditions were determined, different load values applied to the beams and rope were applied. However, within the study, only the results of the applied force of at least 10 tons were shared.

Also, these two conditions were analyzed and evaluated in two ways in order to determine the conditions with and without the load-bearing frame. The analysis reviews are shared in Fig. 4 below.

3. Results and conclusion

For the Beam System, the results were obtained by loading 10 tons within the car load. The highest stress values were obtained both on the rope and on the winch, at the midpoint without the frame. These maximum values were compared with the safety coefficients for the winch and the rope. In conclusion; While the maximum stress value of 329.87 kg/cm^2 is obtained for the rope whose analysis has been completed, it is below the maximum stress value of 333.92 kg/cm^2 for the rope selected according to the TS 1918/14 standard. Likewise, the values obtained when 10tons applied to the beams were applied were 334 MPa for Yield Strength and 513.39 MPa for Tensile Strength.

When the designed system is evaluated according to the data obtained, it is safe. In addition, safer deflection can be achieved by increasing the inertia of the crane beams.

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